Conjecture 00000003420: the spectral formula of effective resistance

Conjecture and reading

"The spectral formula of effective resistance: the resistance is a quadratic form of the pseudoinverse, with the eigendecomposition of the form the harmonic series of eigenvalues."

Conventions. G is any finite connected simple graph (connectedness is a standing hypothesis; without it $e_i - e_j$ need not lie in the range of L), with unit conductances, and $L = D - A$ is Mathlib's `G.lapMatrix` $\mathbb{R}$. The electrical effective resistance between i, j is $v_i - v_j$ for a potential v with $Lv = e_i - e_j$. The pseudoinverse is the Moore–Penrose pseudoinverse. The vague phrase "harmonic series of eigenvalues" is read as the weighted reciprocal-eigenvalue expansion $\sum_{\lambda_k>0}(u_k(i)-u_k(j))^2/\lambda_k$. No other reading is claimed.

Definitions

$L = U\,\mathrm{diag}(\lambda)\,U^T$ is Mathlib's spectral decomposition of the real symmetric matrix L (`Matrix.IsHermitian.spectral_theorem`), U orthogonal with columns the orthonormal eigenvectors u_k. Put $d_k = \lambda_k^{-1}$ if $\lambda_k \neq 0$ and 0 otherwise, and $L^+ = U\,\mathrm{diag}(d)\,U^T$ (Lean: `C3420.pinv`). With $b = e_i - e_j$ (`current`), a potential is v with $Lv = b$ (`IsPotential`).

Theorem

For every finite connected simple graph G: (a) L^+ satisfies the four Penrose equations $LL^+L = L$, $L^+LL^+ = L^+$, $(LL^+)^T = LL^+$, $(L^+L)^T = L^+L$ and is the only matrix that does; (b) for all i, j a potential exists; (c) for every potential v, $v_i - v_j = b^T L^+ b = \sum_{\lambda_k>0}(u_k(i) - u_k(j))^2/\lambda_k$.

Proof

Since $U^TU = UU^T = I$, $LL^+ = U\,\mathrm{diag}(\epsilon)U^T = L^+L$ with $\epsilon_k = [\lambda_k \neq 0]$, and $\lambda\, d\, \lambda = \lambda$, $d\, \lambda\, d = d$ give the four equations (diagonal matrices are symmetric). Uniqueness: if X, Y both satisfy them then $AX = (AYA)X = (AY)^T(AX)^T = (AXAY)^T = (AY)^T = AY$ and likewise $XA = YA$, so $X = XAX = YAX = YAY = Y$.

Existence: $(U^Tb)_k = u_k(i) - u_k(j)$. If $\lambda_k = 0$ then $Lu_k = 0$, so u_k is constant on connected components (Mathlib `lapMatrix_mulVec_eq_zero_iff_forall_reachable`), hence constant since G is connected, so $(U^Tb)_k = 0$. Therefore $\mathrm{diag}(\epsilon)U^Tb = U^Tb$ and $L(L^+b) = U\,\mathrm{diag}(\epsilon)U^Tb = UU^Tb = b$.

Value: if $Lv = b$ then, using $L = L^T$ and $L(L^+b) = b$, $v_i - v_j = v^Tb = v^TL(L^+b) = (Lv)^TL^+b = b^TL^+b$. Finally $b^TL^+b = (U^Tb)^T\mathrm{diag}(d)(U^Tb) = \sum_k d_k(u_k(i) - u_k(j))^2$, and since L is positive semidefinite (Mathlib `posSemidef_lapMatrix`) $\lambda_k \geq 0$, so $d_k \neq 0$ exactly when $\lambda_k > 0$ and then $d_k = 1/\lambda_k$.

Lean

File `lean/Conjecture3420/Basic.lean`. Main theorem `C3420.main : C3420.Claim`, where `Claim := Conjecture` quantifies over all `V : Type` and graphs. Supporting: `moorePenrose`, `moorePenrose_unique`, `LP_current`, `potential_value`, `quad_eq`, `sum_pos`. Neighbouring package 3419 (effective resistance as $v_a - v_b$) was consulted for the electrical definition. Axiom audit: propext, Classical.choice, Quot.sound only (no sorry, no native_decide).